Sustainable Farming Science Program

Inception Report

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List of acronyms

AgMIP	Agricultural Model Intercomparison and Improvement Project
AGRA	Alliance for a Green Revolution in Africa
AI	Artificial Intelligence
AI/ML	Artificial Intelligence and Machine Learning
AoW	Areas of Work
AU-IAPSC	Africa Union – Inter-African Phytosanitary Council
BMZ	Ministry of Economic Cooperation and Development
CIMMYT	International Maize and Wheat Improvement Center
CoPSys	Community of Practice on Systems Analysis
CSISA	Cereal System Initiative for South Asia
DST	Decision Support Tools
EiA	Excellence in Agronomy Initiative
EIAR	Ethiopian Institute of Agricultural Research
FAO	Food and Agricultural Organization of the United Nations
GIZ	German Development Cooperation
HLO	High-Level Output
IAES	Independent Advisory and Evaluation Services
ICAR	Indian Council of Agricultural Research
IFPRI	International Food Policy Research Institute
IMM	Integrated Mycotoxin Management
IPDM	Integrated Pest and Disease Management
IRRI	International Rice Research Institute
ISDC	Independent Science for Development Council
MDII	Multi-dimensional Inclusivity Index
MEL	Monitoring Evaluation and Learning
MELIA	Monitoring, Evaluation, Learning and Impact Assessment
MFS	Sustainable Mixed Farming Systems
NASA	National Aeronautics and Space Administration
NPPO	National Plant Protection Organizations
ODK	Open Data Kit

PAASS	Partnerships for Accelerating Agricultural Solutions at Scale
PAICE	Prioritizing Agronomy in Changing Environments
PHI	Plant Health Initiative
SDG	Sustainable Development Goals
SFP	Sustainable Farming Science Program
ToC	Theory of Change
UAE	United Arab Emirates
VISA	Virtual Institute for Systems Analysis

Executive summary

The Sustainable Farming Science Program (SFP) is designed to address critical demographic, environmental, economic, and social challenges—alongside emerging mega-trends—in Low- and Middle-Income Countries. The program aims to help farmers transform their farming systems to achieve large-scale agronomic gain and whole farm management through an integrated approach focused on women and youth, and combining agronomic, plant health, and farming system solutions. Agronomic gain is defined as enhanced productivity and profitability, improved resource use efficiency (nutrients, water, and labor), healthier soils and crops, and greater resilience to climate change. To maximize impact, SFP works through strategic partnership platforms across prioritized farming systems and countries. Priority regions are South Asia, East and Southern Africa, and West and Central Africa, with additional engagement in Central and Southeast Asia, North Africa, and Latin America.

From 2022 to 2024, three CGIAR Research Initiatives—Excellence in Agronomy (EiA), Plant Health Initiative (PHI), and Sustainable Mixed Farming Systems (MFS)—worked to tackle critical challenges in smallholder agriculture, laying the groundwork for the CGIAR SFP. During the inception phase, the SFP team has elaborated a 2025 work plan to deliver key results of the eight Areas of Work (AoW): AoW 1 advances efforts in generating evidence for climate change adaptation and mitigation on farms; AoW2 focuses on precision soil nutrient management mediated by predictive models and advisories; AoW3 focuses on resilient soil systems, and improved soil health; AoW4 advances plant health and mycotoxin management; AoW5 addresses integrated water management for enhancing water use efficiency; AoW6 focuses on scale-appropriate mechanization; AoW7 advances system-related work; and AoW 8 aims at strengthening MELIA, data and analytics, capacity and policy efforts, and partnership approaches, contributing to program efficiency across all other AoWs.

The knowledge-driven prioritization exercise identified AoW2 (precision soil nutrient management), AoW1 (climate adaptation), and AoW5 (water innovation) as top priorities, with South Asia and Eastern and Southern Africa scoring highest across impact areas. Top-ranked HLOs include nutrient decision support (2.1), prioritization and suitability of water innovations (5.1), eco-friendly pest control (4.4), and soil health monitoring (3.1), though highest-ranking HLOs varied by region. For example, HLO 7.1 (bundled innovations) was highly prioritized for ESA, HLO 5.1 for CWANA, and HLOs under AoW6 (mechanization) for South and Southeast Asia. Enabling HLOs under AoW8—particularly MELIA, data systems, and partnership platforms—also received broad support. While results provide strategic guidance, a second prioritization round will refine direction for the 2026 and 2027 work plans. The SFP team, in consultation with key stakeholders during 2024 and 2025, decided to refocus its AoW structure around key challenges to be addressed through agronomy and farm management R&D, thereby reducing the original eight to five key mission-driven AoWs: (i) Strengthening farm advisory services; (ii) Enhancing preparedness and rapid response to emerging biotic and abiotic shocks; (iii) Resolving farm binding constraints (limiting factors such as soil, water, pests, diseases, and input markets that affect operational efficiencies, profitability, and productivity); (iv) Incubating new transformative R&D initiatives; and (v) Increasing program agility, efficiency, and delivery across all AoWs and Centers, including advancing AI-ready data and data science, strategic prioritization, MELIA, partnership approaches, capacity development and evidence-based policy and investment decisions. The development of the mission-driven Areas of Work has maintained the principles used in writing the SFP proposal. In addition, this has been an inclusive process designed to integrate various disciplines to address key constraints faced at farm level, that are innovative, fit-for-purpose and agile.

1. Program overview

1.1 High-level Vision
SFP is dedicated to delivering innovative solutions that boost farm productivity by an average of 30%, enhance profitability, and improve resource efficiency—reducing production costs by 25%. At the same time, SFP fosters healthier soils and crops and strengthens resilience to climate change, with a projected 15% reduction in greenhouse gas emissions, in interaction with other Programs and Accelerators. Through these efforts, SFP directly contributes to the five CGIAR impact areas. Our mission is to drive agronomic gains in key farming systems where integrated farm management and agronomy can significantly benefit smallholder farmers, particularly women and youth in LMICs. With W1-2 funding alone, we aim to reach at least five million farmers by 2030—a number expected to triple with the inclusion of W3 and bilateral funds.

To scale impact effectively, SFP collaborates through strategic partnership platforms known as Partnerships for Accelerating Agricultural Solutions at Scale (PAASS) within CGIAR regions. The primary focus areas include South Asia, East and Southern Africa, and West and Central Africa, with additional engagement in Central and Southeast Asia, North Africa, and Latin America. PAASS serves as a driving force for agenda-setting, solution development and validation, and scaling initiatives under the AoW framework—ensuring that cutting-edge agricultural advancements reach the farmers who need them most.

SFP is embracing an evolutionary approach, refining its AoW to enhance integration and mission alignment. Starting in 2026, SFP will streamline its original eight discipline-oriented AoW—outlined in the 2024 proposal and implemented during the 2025 transition—into five key mission-driven focus areas, namely 1) Strengthening Farm Advisory Services to leverage CGIAR scientific data to enhance guidance and decision-making for farmers; 2) Enhancing Preparedness & Rapid Response todelivering science-based recommendations to address biotic and abiotic shocks andemphasis on climate change with both short- and long-term strategies; 3) Resolving Key Farm Binding Constraints to tackle soil, water, pest, disease (including mycotoxin), and input market challenges that limit agronomic gain; 4) Incubating Transformative Research & Development with the goal of providing a test space to develop and refine agricultural innovations before broader development; and 5) Increasing Program Agility, Efficiency & Delivery to advance cross-program support activities, including strategic prioritization, MELIA, AI-ready, standardized data and analytics, partnership and capacitation approaches, and evidence-based policy and investment decisions across all AoWs and Centers.

The updated program structure—to be detailed in Q4 2025 as part of the 2026-2027 work plan—will provide a more cohesive and scalable framework for investment, accommodating W1-2, W3, and bilateral funding to advance SFP's core objectives.

1.2 Achievements from 2022—2024
From 2022 to 2024, three CGIAR Research Initiatives—Excellence in Agronomy (EiA), Plant Health Initiative (PHI), and Sustainable Mixed Farming Systems (MFS)—worked to tackle critical challenges in smallholder agriculture through innovation, systems thinking, and strong partnerships, laying the groundwork for the CGIAR SFP.

EiA has played a pivotal role in developing and delivering context-specific agronomic solutions that enhance productivity, profitability, and sustainability for smallholder farmers. Through 20 use cases across 13 countries, EiA has co-developed innovative approaches such as precision nutrient management and scale-appropriate mechanization, directly benefiting 450,000 farmers across 272,000 hectares. Cutting-edge tools —including the AgWise decision support framework, a DataLake, and the Carob data standardization approach — have accelerated the adaptation and implementation of innovations, ensuring their relevance in diverse farming contexts. Beyond direct interventions, EiA has shaped agricultural policy in countries such as Morocco and Vietnam, while also strengthening research efforts in soil health and climate adaptation to drive long-term resilience.

The PHI was dedicated to strengthening global plant health with a focus on LMICs. Its efforts helped enhance pest and disease diagnostics, surveillance, and response capacity, with AI and molecular diagnostic tools deployed across 26 countries to improve early detection and mitigation. By integrating data-driven risk assessment systems and eco-friendly crop protection strategies, PHI has advanced sustainable approaches to plant health management. Additionally, the initiative has played a crucial role in scaling Aflasafe for aflatoxin control, benefiting 11 African countries. Recognizing the importance of inclusive innovation, PHI embedded gender-sensitive approaches in the expansion of digital tools, rigorously evaluating their effectiveness through randomized controlled trials to ensure equitable access and impact.

Implemented in eight countries, MFS employed approaches to develop tools for farming systems analysis, co-design bundled innovations and assess trade-offs and synergies in these systems. Key methodologies address cropping systems diversification, such as crop associations, rotations, including with food-fodder crops, forage seed business development, establishment of fodder grasses in field contours, and farming systems intensification through micro-irrigation and high-value and nutritious fruit trees. Actors from different institutions and disciplines cooperated to develop systemic methods and tools, and context and scale adapted solutions, considering current farming systems and objectives of participating farming communities and a Community of Practice on systems analysis (CoPSys) and the Virtual Institute for Systems Analysis (VISA) are being consolidated with research and education partners.

1.3 Description of intended research

During the inception and transition process in 2025, the Program works through eight AoWs that were approved in the proposal. As the Program's AoW structure is being revised, the research and activities described in Table 1.3.1 refer to the eight AoWs in the first column, with a brief description of the connection with the updated AoWs in the second, which will inform the 2026-2027 work plan preparation.

Table 1.3.1. Summary of current AoW value proposition and intended research and the connections with the future AoW of the SFP.

Intended research in the current AoW structure (2025 work plan)	Connection of the current AoW and the future AoWs*(for 2026-2027 work plan).
The 2025 work plan for **AoW1** prioritizes climate risk research, adaptation and mitigation assessments, stakeholder capacity-building, and informed policy and investment decisions. It begins with the development of high-resolution, spatially explicit tools to map climate hazards and risks across crops and farming systems, enabling precise adaptation targeting. Complementing this, advanced analytics will help farmers prioritize bundled agronomic solutions. A key focus is integrating climate	● Primary corresponding new AoW: AoW2 (Enhancing preparedness and rapid response to emerging shocks) ● Deliverables as inputs into other new AoWs: AoW1 (*Strengthening farm advisory services*), AoW5 (policy evidence, and s*teering agenda setting*)

information services into agronomic advisories, improving farmer decision-making on planting, seed selection, and irrigation through tools like an expanded AgWise recommendation engine. AoW1 will also generate impact assessments on the effectiveness and co-benefits of farming practices, contributing to adaptation tracking and emissions reduction evaluations. To support adoption, the program will invest in capacity-sharing innovations and gender-responsive pathways to scale, with targeted support for marginalized groups. Together, these activities provide a coherent research agenda that links risk, response, and readiness, driving climate resilience and sustainability across farming communities.	• Expertise relevant to other AoWs: AoW4 (*Incubating transformative agricultural R&D*) Current AoW1 activities will be connected primarily with the new AoW2 through support for the development of climate-resilient farming systems.
By the end of 2025, **AoW2** will start delivering products such as databases on nutrient removal, organic inputs, biofortification, and compiled cocoa/coffee datasets; frameworks integrating nutrient and acid soils management, and seasonal forecasting into the AgWise decision support framework and recommendation engine; AI-based models for canopy imaging and crop monitoring; and enhanced farmer-oriented advisory tools. The AoW will generate new fertilizer and lime recommendations for five or more countries, and validate them through on-farm trials to advance work started under EiA. Additionally, training materials to implement AgWise and other interventions will support farmers and scientists in national programs including for Egypt, Rwanda, DRC, and Kenya. To further enhance equity and impact, AoW2 will also produce a report on gender-related constraints for Nigeria and launch a web-based dashboard on agronomic biofortification, ensuring accessibility to critical insights for informed decision-making.	• Primary corresponding new AoW: AoW1 (*Strengthening farm advisory services*), AoW 3 (Resolving farm binding constraints) • Deliverables as inputs into other thematic areas: AoW5 (policy evidence, and steering agenda setting) • Expertise relevant to other new AoW: AoW4 (Incubating transformative agricultural R&D) Current AoW2 activities around developing and delivering tailored fertilizer recommendations are connected to agricultural advisory activities in the new AoW1 and to addressing farm-level constraints in AoW 3.
During 2025, **AoW3** will start implementing a soil health monitoring framework with validated standard operating procedures, integrating biological and molecular indicators alongside advanced digital tools for data analytics and AI. Long-term monitoring sites across diverse agro-ecosystems will be mapped to track carbon storage, biodiversity, and hydrologic functions, while the impacts of agronomic practices on soil resilience will be quantified to guide sustainable investment decisions in priority regions. To support inclusive, data-driven decision-making, evidence-based insights on soil health will be translated into digital tools and policy briefs, incorporating gender-transformative approaches that enhance soil health awareness and capacity building.	• Primary corresponding new AoW: AoW3 (Resolving farm binding constraints), and AoW4 (Incubating transformative agricultural R&D) • Deliverables as inputs into the new AoW5 (policy evidence, and steering agenda setting), and AoW1 (*Strengthening farm advisory services*). Current AoW3 activities on resilient soils will continue addressing farm-level constraints in the new AoW3.

In 2025, **AoW4** is focused on tackling biotic constraints by strengthening plant health diagnostics and surveillance networks, enhancing data management and modeling tools, and implementing risk anticipation and prevention strategies through policy briefs for decision-makers. The AoW works across Centers and partners to scale validated eco-friendly, inclusive, and affordable Integrated Pest and Disease Management (IPDM) and Integrated Mycotoxin Management (IMM) solutions, while developing cost-effective traceability systems for mycotoxins and pesticide residues to improve food safety. All centers, along with several partners in LMICs, are actively engaged, addressing key pests and diseases of economic importance in specific sub-regions. Efforts include modeling pest outbreaks, mapping hotspots, piloting remote sensing, ICTs, and AI-driven methods for biotic stress detection, as well as evaluating nano sensor-based innovations, IPDM, and nature-based solutions for pest, disease, and mycotoxin control. Ultimately, AoW4 aims to deliver validated disease management protocols for key food crops to facilitate inclusive scaling, while also strengthening biosecurity, seed health surveillance, and digitizing training resources for plant protection experts.	● Primary corresponding new AoW: AoW2: (Enhancing preparedness and rapid response to emerging shocks), and AoW3 (Resolving farm binding constraints) ● Deliverables as inputs into the new AoW1 (*Strengthening farm advisory services*), and AoW5 policy evidence, and steering agenda setting) ● Expertise relevant to the new AoW4 (Incubating transformative agricultural R&D) Current AoW4 activities will be connected primarily with the new AoW2 through strengthening prediction, preparedness and response systems, and the new AoW3 for deliverables related to building resilience against biotic binding constraints.
During 2025, **AoW5** is structured around three core components: understanding water-related risks in agriculture, developing integrated water management practices for rainfed, supplementary, and irrigated systems, and leveraging evidence to create training materials and inform policies. In the first six months, efforts focused on finalizing ongoing initiatives, including establishing a cross-center framework for water-related data generation, spatial prioritization, and KPI measurements from field and farm-level interventions. Additionally, AoW5 has played a key role in global dialogues on water innovations, contributing to CGIAR Water Week, the Water for Food Global Conference in Nebraska, and supporting the newly formed Global Commission on Water for Food Futures, helping to shape the future of sustainable water management in agriculture.	● Primary corresponding new AoW: AoW3 (Resolving farm binding constraints) ● Deliverables as inputs into the new AoW1 (*Strengthening farm advisory services*), and AoW5 (policy evidence and steering agenda setting) ● Expertise relevant to the new AoW4 (Incubating transformative agricultural R&D) The current AoW5 activities will be connected primarily with the new AoW3, addressing farm-level biophysical constraints around water management.
AoW6 recognizes farm mechanization as a crucial enabler for implementing effective crop management practices amid production constraints. This AoW overcomes socio-economic barriers and diverse agroecological conditions in the Global South by synthesizing fragmented lessons from various contexts, providing a structured approach to identifying, adapting, developing, and documenting viable	● Primary corresponding new AoW: AoW3 (Resolving farm binding constraints) and AoW1 (*Strengthening farm advisory services*, with content related to mechanization)

technology bundles and delivery models, ensuring scalable mechanization solutions that align with sustainable intensification principles. In 2025, AoW6 will introduce a global scale-appropriate mechanization (SAM) dashboard detailing country-level mechanization status, conduct a mechanization gap analysis for key cropping systems, and develop geospatial mechanization suitability maps. Additionally, AoW6 will produce two global knowledge products: a SAM Options Inventory, encompassing field and post-harvest tools, and a Service Provision Models Inventory, offering strategies for effective mechanization delivery.	• Expertise relevant to the new AoW4 (incubating transformative agricultural R&D) AoW6 activities will be connected primarily the new AoW 3 and 1 around machinery related options and services.
AoW7 plays a critical role in SFP by bridging research from other AoWs employing a whole-farm perspective to integrate solutions at farm level. Implementation is via scalable socio-technical innovation bundles co-designed and co-created with partner organizations and farmers. These combine on-farm technologies and practices with enabling services and policy measures. In 2025, AoW7 initiated collaborative research with partners and other AoWs and Programs to develop farm-level innovations, including crops, livestock, trees, and fish, ensuring their integration within specific contexts while fostering an enabling environment for adoption. Additionally, efforts were made to support institutional transformation, including within CGIAR, by advancing co-creation methodologies and systems science approaches through capacity-sharing initiatives such as the Virtual Institute for Systems Analysis (VISA) to further strengthen systems-based agricultural innovation.	• Primary corresponding new AoW: AoW3 (Resolving farm binding constraints), and AoW1 (*Strengthening farm advisory services*) for assessing recommendation adoption and trade-offs. • Expertise relevant to all other new AoW: Current AoW7 approaches on system integration will contribute to all the new AoW and future solutions to be assessed with farming system lenses, particularly emphasizing response to farmer (women and youth) demands. While the current AoW7 will contribute to all new thematic areas, the most direct connection will be with activities around designing bundles of solutions to binding constraints, and farm advisory services.
AoW8 plays a crucial integrative role within the SFP, supporting cross-cutting functions across all AoWs through agenda setting via prioritization approaches, MELIA, data standardization, analytics support, capacity sharing, partnership approaches (e.g., PAASS), and policy advice. During 2025, these core activity clusters are working to develop a logic to prioritize farming systems for agronomic gains, supported by a digitally enabled MELIA system for large-scale assessments; improve infrastructure to share, standardize and advance AI-ready, FAIR data and analytics (with the Digital Transformation Accelerator); design capacity-sharing models and tailored training programs for national scientists and last-mile staff (with the Capacity	• Primary corresponding new AoW: AoW 5 (data and analytics, MELIA, capacity sharing, policy advice and agenda setting) • Other current AoWs as an input into the new AoW5: All AoWs will contribute evidence and learnings into the new AoW5 and represent a testing ground for new developments and tools from it. • AoW8 expertise and deliverables will be inputs into all new AoWs. AoW8 products and approaches will provide tools, models, and other

Sharing Accelerator); develop PAASS logic and guidelines for co-designing and scaling bundled solutions (with the Scaling for Impact Program); and synthesize evidence on scalable bundled solutions and enabling conditions to inform policies on subsidies, soil and water management, and climate resilience.	infrastructure support for the SFP's efforts, including on data and analytics, MELIA, capacity sharing, and policy and agenda setting. AoW8 corresponds to the new AoW5.

* The proposed new AoW will be interconnected. For example, once scientific evidence on future shocks (AoW2) or binding constraints (AoW3) is developed, that evidence will feed the AI based advisory models in the new AoW1. Similarly, the new AoW4 (oriented to incubate new and blue-sky research ideas) will contribute to the other AoW if the new ideas have the potential to evolve into concrete results and solutions.

2. Codesign and partnerships

SFP's activities are developed through co-design processes with a diverse range of partners, informed by extensive stakeholder consultations. During the Initiative implementation, demand-driven agronomy and agricultural solutions were advanced through engagement approaches, including close collaboration with demand and scaling partners. To enhance partnership engagement, SFP's three constituent Initiatives (EiA, PHI and MFA) co-developed Partnerships for Accelerating Agricultural Solutions at Scale (PAASS)—multi-partner scaling platforms tailored to priority geographies and farming systems. These platforms identify critical biophysical and socio-economic constraints and biotic and abiotic shocks and co-create targeted solutions to improve farm advisory services. To guide implementation, a PAASS playbook was developed, outlining its principles, objectives, structure, core functions, partner roles, and setup processes. Ultimately, PAASS serves as a key scaling mechanism for agricultural innovations, while keeping other partnership models open for further research and development.
The Prioritizing Agronomy in Changing Environments (PAICE) framework, developed with partners (including CSIRO and Cornell University) under the aegis of EiA, was designed to identify priorities for climate change adaptation and sustainable intensification through participatory and data-driven approaches. Concluded in late 2024, it engaged 485 participants from 216 institutions, covering 75 million hectares across 12 production systems (see description and summary results). Its insights have been used to compare high-level outcomes (HLOs) across AoW, inform donor discussions on future investments in SFP at both global and AoW levels, and refine the program's operational model and prioritization within PAASS. Additionally, PAICE findings are shaping the restructuring of AoW, ensuring a more targeted and effective approach to sustainable agriculture.

In India, the Cereal Systems Initiative for South Asia (CSISA) team (IFPRI, IRRI, and CIMMYT) collaborated with public and private partners, including the Gates Foundation, to assess gaps in digital advisory services and explore CGIAR's potential contributions. During the Digital Agriculture Conclave on May 8 in Delhi, leading tech firms conducted a landscape analysis to identify key areas for improvement. CSISA, in partnership with ICAR, identified four strategic advantages CGIAR could bring to AI-enabled farm advisories: closing big data gaps by enabling cost-effective, AI-driven data collection on farming practices at scale; enhancing recommendation logic through AI/ML advancements that address landscape complexity and farmers' constraints; researching adoption barriers, including behavioral, social, and economic factors, to improve uptake; and developing demand-driven advisories that ensure more responsive and farmer-focused recommendations. These insights will help refine digital agriculture strategies for more effective scaling and impact.

Partner consultations in Africa have identified priorities including AI-driven agro-advisories (e.g., with Digital Green), fertilizer recommendation validation using harmonized decision support tools (e.g., in Ethiopia), and lime-recommendation development such as that coordinated by EIAR. Additionally, a consultation workshop held from May 7–9, 2025, in Nairobi, Kenya, assessed country capacities and gaps in managing banana bunchy top disease. Organized with the Inter-African Phytosanitary Council and FAO, the workshop brought together national plant protection agency (NPPO) leaders from Benin, Burundi, Kenya, Malawi, Nigeria, Rwanda, Tanzania, and Uganda, fostering collaboration to strengthen regional plant health strategies.

The SFP team has engaged funders and key stakeholders in ongoing discussions to optimize the Program's AoW structure for maximal impact. During meetings in Nairobi (March & April 2025) and Ghent (May 2025), the team agreed to transition from disciplinary AoW to a more integrative, results-driven areas (see Section 1.1). Stakeholders reinforced SFP's comparative advantage for supporting data-driven advisory services for farming solutions, identifying it as a key future AoW. To strengthen its position in the digital landscape, SFP will dedicate an entire AoW to advancing AI and scientific evidence for scalable, high-impact agronomic farm advisories in close collaboration with experienced partners on the subject.

At a global level, the SFP engaged stakeholders from academia, the private sector, and public institutions during the Water for Food Global Conference in Nebraska (April 28 – May 2, 2025). A consultation hosted by the Commission on Water for Food Futures reinforced the importance of the future SFP AoWs, highlighting key challenges such as managing agricultural production amid climate uncertainty, alternative water sources and reuse for food production (including addressing salinity issues), drainage as a crucial component of agricultural water infrastructure, and integrated water management solutions, including drought-resistant crops paired with optimized water practices. These insights will help shape SFP's strategic approach to agricultural water management.

3. Theory of change and MELIA

3.1 Theory of change (ToC)

The SFP addresses pressing demographic, climatic, environmental, and social challenges, focusing on feeding a growing population, resource conservation, and equitable farming opportunities, particularly for women and youth. The Program aims to contribute to the five CGIAR Impact Areas by co-developing and scaling whole-farm solutions that enhance productivity, soil health, resource efficiency, and climate resilience, while reducing crop losses and environmental harm. Sustainability—spanning profitability, environmental benefits at farm level, GHG emissions' mitigation and climate resilience, and social equity—is central to its approach. Through collaboration with CGIAR Programs, Accelerators, and partners, SFP expects to support at least five million farmers across two million hectares (via pooled funding), increasing this figure three-fold through W3 and bilateral projects. Farm productivity is expected to increase by 30% by scaling SFP innovations, and production costs and GHG emissions to reduce by 25 and 15%, respectively. Additionally, the Program will strengthen national research systems by building capacity among 25+ national science partners and improving the sharing of tools, data, and methods across CGIAR Centers to inform future pooled and bilateral projects.

The HLOs will be achieved through three interconnected pathways: innovation, capacity sharing, and policy change. The innovation pathway prioritizes contextualized solutions, integrating climate adaptation, nutrient management, soil health, pest, disease and weed (PDW) control, water management, mechanization, and system integration, supported by participatory design, standardized, available data and advanced analytics. Gender- and youth-inclusive innovation bundles will be based on a growing corpus of increasingly human and machine-actionable data, AI and prioritization tools, largely implemented via PAASS platforms. These efforts will be strengthened through strategic partnerships, (e.g., with GIZ/BMZ, AGRA, the Sustainable Agriculture Initiative, the Sustainable Intensification Innovation Lab, and key funders like the UAE and Gates Foundation). The capacity sharing pathway focuses on delivering tailored training materials for farmers and extension workers, supporting partner use and sharing of data and associated tools, to enhance NARES capabilities. The policy change pathway leverages evidence, policy briefs, multi-stakeholder platforms, and international dialogues to drive investment and adoption of farm innovations at scale.

In 2025, HLOs are anchored in the eight AoWs included in the SFP proposal. Most AoWs emphasize co-developing solutions within specific farming systems and geographies, guided by the three interconnected pathways, while others focus on system integration and program-wide delivery. AoW8 plays a cross-cutting role, supporting other AoWs and aligning Programs and Accelerators through essential functions such as prioritization, analytics, policy engagement, and MELIA, ensuring a cohesive, evidence-based and impactful approach to agricultural innovation.

The SFP ToC relies on key assumptions across three spheres. Within the sphere of control, it assumes sustained financial, human, and technological resources, effective contributions to and functioning of data systems and digital tools, and stakeholder capacity and willingness to engage. In the sphere of influence, it presumes that scaling partners and farmers will adopt and expand innovations, feedback mechanisms will enhance delivery, and institutional actors will align with evidence to shape policy environments. Finally, within the sphere of interest, it is expected that innovations can be scaled towards impact, that behavior change will persist beyond direct program support, and that favorable market, policy, and climatic conditions will enable long-term and sustainable improvements in productivity, resilience, and equity. *Open Access link for both the ToC and the Results Framework.* *(Zoom out to see the linkages of the various AoW ToCs and download the ToC tables using the 4th icon on the top left conner of the page).*

Changes made

As indicated in Section 1.1, the SFP Leadership Team has agreed to restructure its AoWs, transitioning from the existing eight disciplinary AoWs to a more results-oriented framework (staring in 2026) with five key focus areas: strengthening farm advisory services, enhancing preparedness and rapid response to biotic and abiotic shocks, addressing critical farm binding constraints (soil, water, pests, diseases, and input markets), incubating transformative research and development (R&D) initiatives, and increasing program agility and efficiency. This shift will require adjustments to the Program-level ToC, including updates to AoW ToCs, results statements, and performance indicators. The revised ToCs, narratives, results framework, and assumptions will be finalized in Q4 2025 as part of the 2026-2027 work plan.

Considering this evolution, the Inception Phase introduced key changes to the ToC, incorporating new assumptions and a research question assessing the role and types of data needed for farm advisory services. Revisions to results statements included removing embedded targets and making innovations—such as farm advisories, data, and analytics—more explicit. The ISDC feedback on the Program-level ToC and AoWs was addressed (see Appendix 1), refining the definition of Farming Systems and integrating relevant indicators into corresponding outcomes. Assumptions were linked to both narratives and ToC diagrams, ensuring alignment with the result framework and MELIA studies for ongoing monitoring. Innovations were outlined across all three results levels, with a focus on advisory services and technologies to support preparedness to shocks and resolving farm binding constraints such as climate-smart practices, nutrient management, soil and water management, plant health protection, mechanization, and integrated farm solutions, all underpinned by data and analytics. The ToC reinforces the Program's broader system positioning, ensuring AoWs are linked through co-development platforms while fostering collaboration with other programs and accelerators during work planning. Additionally, mapping of 61 bilateral projects to the eight AoWs confirmed that each AoW is connected to at least two projects with direct result contributions.

3.2 Monitoring, Evaluation, Learning, and Impact Assessment (MELIA)

The MELIA plan provides a comprehensive framework for tracking SFP results, from HLOs to 2030 Outcomes, while assessing contributions to CGIAR Impact Areas and SDGs. It integrates aligned indicators and targets, with data collected across all levels and AoWs through a digitally enabled system that supports near-real-time tracking and facilitates data integration on innovation development, capacity sharing, scaling, and use.

For Monitoring, Evaluation, and Learning (MEL), output-level data—such as number of people trained, innovations co-developed, and scaled—will be tracked using standardized Open Data Kit (ODK) tools across all AoWs, managed by PAASS platforms, AoW leads, and implementing partners. The MEL team, in collaboration with the AoW leads, PAASS coordinators and Capacity Sharing Accelerator, will ensure key variables are collected and tools adapted as needed. Data-sharing partners with existing systems will connect via an Application Programming Interface (API), while those without will receive capacity-building support. Additionally, knowledge and innovation products will be monitored through a shared repository for annual reporting at the AoW level.

In addition to the overall MELIA framework covering all AoWs, the MELIA team will work with each AoW to align data needs, integrate tools, and develop specific MEL strategies where applicable. This ensures consistency in data quality, timely feedback, and contributions from all AoWs.

To track outcome-level data, the MEL team will conduct annual household and partner surveys, key informant interviews, and focus group discussions, complemented with bi-annual phone surveys capturing innovation uptake and real-time feedback. In 2025, the digital Multi-Dimensional Inclusivity Index (MDII) tool will be piloted across select digital advisory tools, assessing gender and social inclusivity while informing recommendations for development teams. The MEL team will determine future MDII applications based on pilot results, evaluating digital inclusivity in farm advisory services, refining enhancement strategies, and tracking effectiveness. AI approaches including chatbot will be explored to gather real-time user feedback, while baseline and partner feedback surveys will establish benchmarks, assess engagement quality, and evaluate the use of research outputs in decision-making.

The updated framework addresses previous evaluation findings, particularly the limited application of learnings across Centers due to varying systems and frameworks. To enhance standardization and comparability, common standards and data collection tools will be implemented, ensuring consistent data and results across participating Centers and partners. Additionally, a mid-term evaluation in 2027, led by the Independent Advisory and Evaluation Services (IAES), is planned to assess Program impact. This evaluation will be funded by IAES, with the MELIA team providing necessary data and reporting support.

For impact assessment, the Program will conduct MELIA studies across diverse geographies to generate evidence on innovation performance, adoption, and impact. Key indicators include crop and animal productivity, profitability, resource use efficiency, and soil health metrics such as soil organic carbon stocks, carbon balance, and climate change mitigation potential (GHG emissions).

The planned studies include Causal Impact Assessment learning studies on mechanized direct-seeded rice in Cambodia and Vietnam, site-specific agronomy in Ethiopia, and planting date advisories in India. Additionally, public-private extension models in Rwanda and institutional support for sustainable intensification in Ghana and Malawi will be evaluated. Broader adoption and impact studies, along with a Qualitative Outcome study on gender-focused research, will be conducted in Ethiopia, Ghana, and Malawi. In Nigeria and Uganda, research will assess the impact of Aflasafe promotion among maize farmers using subsidies and awareness campaigns and push-pull technologies (biological pest and weed control technology) adoption respectively, while further studies in Zambia, Nigeria, Senegal, Ethiopia, and Mali will assess adoption and success of digital extension tools, and good agricultural practices. The team will coordinate joint implementation at the country level, with center-level budgets allocated to support the 2025 studies, many of which build on legacy initiatives and contribute to evidence-based SFP results. There will be collaboration with other CGIAR Programs and Accelerators such as Climate action, Scaling for Impact, and Gender Accelerator to align evidence generation, support joint learning, and enhance the design, uptake, and scaling of innovations across shared priority areas. Please refer to the MELIA plan and the results framework attachments.

4. Comparative advantage analysis

The comparative advantage analysis identified a diverse set of actors, including academic and research consortia (e.g., Wageningen University & Research, AgMIP, NASA Harvest), national agricultural research institutes (e.g., ICAR, EIAR), government bodies, non-governmental organizations (e.g., Digital Green), and private sector entities. This assessment allowed SFP to evaluate its role across 34 High-Level Outputs (HLOs), highlighting its strengths in systems-based, long-term, and multi-disciplinary research, while identifying areas where other organizations are better suited for implementation, digital delivery, or localized contextualization. Mapping this landscape enables the Program to avoid duplication, address gaps CGIAR is uniquely positioned to fill, and establish complementary roles with partners. Additionally, the analysis supports strategic prioritization, helping define where CGIAR should lead, co-lead, or support, ensuring improved coordination, targeted investments, and a stronger contribution to global food system sustainability. Please refer to the comparative advantage analysis attachment.

Across the 34 HLOs analyzed, the Program is well positioned in 65%, moderately positioned in 32%, and weakly positioned in 3% (see column I in the link above). However, even in areas where SFP has a strong presence, partnerships remain essential for successful adoption and impact, and to ensure that innovations are scaled, contextualized, and aligned with stakeholder needs.

For HLOs where CGIAR is moderately positioned—including 1.1, 1.2, 1.4, 2.3, 2.4, 3.2, 4.2, and all four HLOs in AoW7—the Program recognizes the need for strong collaboration with well-placed actors such as advanced research institutes, national institutions, digital platforms, and policy-focused organizations. These partnerships ensure effective implementation, local contextualization, and policy alignment. Even in areas where CGIAR has a strong presence, partnerships are critical for scaling innovations, adapting solutions to specific contexts, and amplifying policy influence.

To maximize impact, the Program will prioritize co-design, joint implementation, and knowledge exchange, reinforcing a collaborative approach that strengthens the research-to-impact continuum and enhances global food systems sustainability.

For HLO 1.3, which focuses on developing training materials to strengthen stakeholder capacity, CGIAR's comparative advantage is limited compared to specialized digital extension and outreach organizations. However, the Program remains actively engaged, recognizing that CGIAR's expertise in technical farming systems knowledge (HLOs 1.1 and 1.2) must be effectively integrated with modern training methodologies. Without demand-driven, accessible, and context-specific training tools, the uptake of CGIAR's scientific outputs would be significantly constrained. To address this, the Program will collaborate closely with digital platforms and national extension systems to co-develop materials, while innovating new ways to enhance last-mile delivery. The priority is to create locally tailored, responsive content, ensuring that scientific advancements translate into actionable, real-world solutions for farmers and other stakeholders.

As the SFP undergoes restructuring and establishes new AoWs, updating the comparative advantage analysis will be crucial. This reassessment will help the Program refine its strategic focus, strengthen partnerships, and optimize resource allocation within the broader ecosystem, ensuring alignment with emerging priorities and operational goals.

5. Prioritization

This section provides the results of the knowledge-based standard prioritization methodology used across Programs and Accelerators, which is included in the link below.

5.1 Knowledge-driven priorities

The knowledge-driven prioritization exercise supports SFP in decision making about priority HLOs, AoWs, and regions. The process began with the development of justification by a core team, leveraging large-language model technology (i.e., Claude, Gemini, and ChatGPT), to enhance consistency and coverage. AoW leaders then reviewed and refined scores and justifications, including capping HLOs for impact areas with primarily indirect effect (e.g., climate change) and boosting region-specific scores where expert judgement indicated greater relevance. Final scores were rebalanced to comply with the max allocation rules. Please refer to the prioritization attachment.

Overall Findings
Across all impact areas and regions, AoW2: Precision Nutrient Management received the highest average prioritization score (mean = 0.29, SD = 0.17), closely followed by AoW1: Climate Adaptation and Mitigation (mean = 0.28, SD = 0.17). AoW2 was particularly valued for its role in enhancing agronomic gain, optimizing spatial targeting, and improving nutrient use efficiency. In contrast, AoW1 served as a foundation for risk-informed planning and adaptation investment. However, while AoW2 was the highest average score, the top priority AoW varied by region highlighting the importance of regionally targeted programming.

Among impact areas, the program is expected to contribute most to Climate Adaptation (mean = 0.26, SD = 0.14), followed by Environmental Health and Food Security (both at 0.25). These scores may reflect the alignment of AoWs with high-profile global goals and external demand, particularly in climate responses. However, the similarity between climate and environment may partly result from methodological overlap, as both impact areas used similar scaling indicators. Poverty

Reduction and Social Inclusion scored only slightly lower (mean = 0.23), reinforcing the SFP's importance across all five impact areas.

Regional Prioritization

Analysis revealed large regional differences: South Asia (SA) ranked highest (mean = 0.38, SD = 0.15), followed by Eastern and Southern Africa (ESA) (mean = 0.31, SD = 0.14), aligning with region-specific justification and performance across multiple AoWs. SA scored highly in nutrient management (AoW2), mechanization (AoW6), and water innovation (AoW5), reflecting policy momentum and scalability. ESA performed strongly in pest and disease management (AoW4), climate information systems (AoW1), and digital decision support (AoW2). Latin America and the Caribbean (LAC) showed concentrated strengths in water and digital systems (AoWs 5 and 8), while Central and West Asia and Northern Africa (CWANA) performed notably in AoWs 1 (climate) and 5 (water). In contrast, Southeast Asia (SEA) (mean = 0.22) and West and Central Africa (WCA) (mean = 0.21) were prioritized in fewer HLOs—an outcome that is difficult to reconcile with the known challenges and needs in these regions.

AoW × Region Interaction

The prioritization revealed high variability within and among AoWs across regions, demonstrating that prioritization is influenced not only by individual regions or AoWs but also by their intersection. For example:

- AoW4 (Pest/Disease Management) is highly prioritized in ESA due to elevated biotic stress.
- AoW5 (Water Management) ranks highly in WCA and CWANA due to limited irrigation access and water insecurity, as well as in SA, where rainfed system vulnerabilities and excessive irrigation use drive prioritization.
- AoW6 (Mechanization) is emphasized for the development of global knowledge products, drawing insights from key regions across continents, particularly ESA, SEA, and LAC, where youth unemployment and labor shortages constrain productivity.

Furthermore, when prioritization is analyzed beyond the AoW level, the importance of HLOs within AoWs varies strongly across regions. However, many HLOs are not independent of one another—several are co-dependent and mutually reinforcing, contributing together to the achievement of outcomes at the impact area level. This interdependence complicates efforts to prioritize HLs in isolation and underscores the need for integrated, geographically nuanced research deployment at the AoW and HLO levels.

Prioritized HLOs

The highest scoring HLOs across regions and impact areas were:

1. HLO 2.1 – Spatially-explicit nutrient decision-support systems (mean = 0.35).
2. HLO 5.1 – Prioritization and suitability of water innovations (mean = 0.32).
3. HLO 4.4 – Eco-friendly pest management (mean = 0.32).
4. HLO 1.1 - Quantified climate hazards; 1.2 - Climate-specific innovations; 1.4 - Policy and investment-ready information on sustainable farming solutions; 2.2 – Expanded data-driven agronomic recommendations (mean = 0.29).
5. HLO 3.1 – Soil health monitoring and assessment framework (mean = 0.28).

These HLOs offer high-leverage entry points with clear pathways to impact and strong regional relevance. HLO 2.1 enhances spatially targeted agronomic decision-making, particularly in ESA and SA. HLO 5.1 supports climate-resilient water innovation, targeting both water-scarce regions and areas where agricultural water use is excessive. Similarly, HLO 3.1 ranked consistently high across all five impact areas (mean = 0.28), reinforcing its critical role in productivity, resilience, and environmental accountability. Several additional HLOs scored comparably, though slightly lower. Given the limitations of subjective scoring and methodological trade-offs, it may be difficult to meaningfully distinguish between these in a ranked list. As with AoW-level prioritization, top HLOs varied by region, reinforcing the importance of region-specific targeting based on positive outliers rather than relying solely on global averages.

Implications for Targeting

Prioritization findings would support prioritizing investment in AoWs 2, 1, and 5, particularly in SA and ESA, where high regional scores, scientific justification, and institutional demand reinforce their significance (see Section 2: Co-design and partnerships). At the same time, AoWs that indirectly support program implementation, such as AoW8, received broad support. However, knowledge-driven prioritization provides a relatively weak foundation for portfolio design, funding alignment, and programmatic focus. Several limitations—such as potential bias and fatigue in expert elicitation, suboptimal indicator selection (e.g., metrics that inadequately capture key changes, such as greenhouse gas emissions for climate change adaptation), and scale mismatches between regional scores and subnational work—undermine its applicability. Future iterations will need to adopt a clearer logic, a more flexible structure, and stronger internal validity to increase relevance and decision utility.

5.2 Final decisions on the Program's direction and design

Despite the challenges inherent in this approach, results from this prioritization exercise can inform decisions on the strategic direction of the Sustainable Farming Program but results must be interpreted with caution given the methodological caveats. Still, the final configuration of the 2025 Program reflects four key design principles:

1. **Investment in High-Priority AoWs**:
   For future research, SFP should prioritize topics and solutions related to AoW2 (precision nutrient management), AoW1 (climate adaptation), and AoW5 (water innovations), which consistently received high scores across impact areas and regions. AoW4 will also remain a priority, given the ongoing and emerging threats posed by pests and diseases to global crop yields, as highlighted in the PAICE report. These AoWs provide direct, measurable outcomes and align closely with CGIAR's comparative advantage in advancing sustainable farming.

2. **Geographically Differentiated Implementation**:
   South Asia and Eastern and Southern Africa should be the primary geographies for intervention, given their high prioritization scores, strong institutional demand, and potential for system transformation. Although WCA, LAC, and CWANA may have been deprioritized due to methodological factors, the Program will take a more targeted approach in these regions, focusing on AoWs and HLOs that offer the strongest local fit.

3. **Focused Support to Enabling Systems**:
   Enabling functions (AoW8) remain critical, and early-stage investments will lay the foundation for effective implementation. These functions, which include MELIA, data and analytics, and partnership platform approaches facilitating demand articulation, implementation and scaling—will be embedded in delivery mechanisms where readiness and demand are clear.

4. **Accommodations for budget reductions, examples of activities to be stopped:**
   AoW1 has refocused activities and stopped site specific adaptation and mitigation activities, while increasing focus on regional and farming system activities.
   Similarly AoW2 prioritized activities that can contribute to models and advisories, and reduced those that do not contribute to this goal.
   Regarding AoW 3, the activities that have been paused involve the mapping and cataloging of long-term experiments and monitoring sites across the prioritized regions. Additionally, the feasibility study including documentation of assessing the integration of existing C-stock models (model ensembles) to enhance modeling capacity for diverse land use and geographic contexts has been stopped.
   AoW4 will reduce surveillance activities, development of detection and diagnostic tools and validation of IDPM for certain pests and diseases entirely or in certain regions. There will also be a reduction in capacity building activities, funding going to partners and mycotoxin management. Indicative examples of deprioritized activities include discontinuation of late blight and cassava common mosaic virus surveillance, development of remote sensing-based disease severity estimation in rice, beans and other crops, piloting and scaling of tomato and legume IPDM approaches and further development of DSS tools for several diseases. Further development of diagnostic tools for novel potato viruses and cereal rust will be postponed as will studies on major forage pest & Mycotoxin management on feed, feed contamination, implication, and management.
   For AoW 5, activities that have been reduced are those related to the identification of water related risks to and from agricultural systems, and the testing of water innovations in rainfed and irrigated systems. The work related to informing policy and investments have been put on hold.
   AoW7 reprioritization implied suspending activities that need physical experimentation and field validation and further data generation or collection. During 2025, AoW7 is focused on activities that can be addressed at desktop level to produce knowledge outputs from the analysis of data generated during early 2025 and from MFS Initiative activities.
   AoW8 has deprioritized some high-level deliverables under almost every HLO, along with activities that were initially meant to address all SFP themes. These deprioritized deliverables and activities will be added in subsequent years as the budget allows. Indicative examples of deprioritization for HLOs is as follows. For 8.1 (MELIA, ex-ante/prioritization), revision of an ex-ante assessment and investment targeting tool for fertilizer and soil health investments; some MELIA baseline studies) activities. For 8.2 (data and analytics), improvements to the impact dashboard; reduced data standardization targets. For 8.3 (capacity), the number of trainings and the number of MS/PhDs will be reduced. For 8.4 (multi-stakeholder partnership platforms), the number of platforms for which a packaging and scaling strategy is formalized; integrative solutions for scaling; and the number of learning/planning workshops will be reduced.

These decisions reinforce a lean, impact-driven program structure that balances scientific ambition with operational pragmatism, considering W1-2, W3 and bilateral investments.

Prioritization provides a foundation for internal coordination and external communication, though further refinement is needed to fully achieve its intended goals. The results will inform AoWs and regional reorientation during the next planning phase; **Prioritization v2.0** will be implemented to guide efforts from 2026 to 2030.

6. Alignment of W3 and bilaterally funded work

For 2025, the SFP has mapped 61 W3/Bilateral projects from participating Centers, collectively expected to contribute approximately $80 million over the next five years, (translating to $46m for 2025). This investment supports SFP's high-level objectives and advances CGIAR impact areas. Through these projects, Center teams have strengthened partnerships and collaboration while diversifying funding sources, securing contributions from over 20 funders—including private sector foundations (e.g., Gates Foundation), national and federal governments, ministries, government agencies, and NGOs (e.g., IFDC). W1-2, W3 and bilateral funding enables an expanded geographic reach, covering all CGIAR regions. Among the 12 centers engaged in SFP, 10 have bilateral and W3 projects closely aligned with its AoWs, including climate change, precision nutrient management, resilient soils, plant health, scale-appropriate mechanization, water management, and mixed farming systems. Additionally, these projects prioritize strategic crops such as banana, cassava, sorghum, wheat, maize, potato, and millet, with some initiatives specifically addressing the needs of women, youth, and marginalized groups, including internally displaced people and refugees. Below, we illustrate how three bilateral projects—each allocating approximately 30% of their budgets—are aligned with and contribute to SFP's high-level objectives. These projects advance SFP's AoW2 by supporting site-specific nutrient recommendations to increase yield and profitability, AoW3 by validating intercropping-based nutrient strategies that enhance soil health, and AoW5 by supporting water use efficiency.

i. Project iSPARK (Donor: FCDO; Lead: ABC; End: 31/12/2026; SFP contribution: $406,594) promotes sustainable farming in Kenya by integrating innovation and metrics related to sustainability, resilience and adaptation into agro-advisories and policies, supporting data-driven, climate-resilient decision making.
ii. Project Fertilizer Rights Vietnam (Donor: USDA; Lead: IRRI; End: 31/03/2027; SFP contribution: $235,594) enhances rice fertilizer-use efficiency and effectiveness through site-specific nutrient management and low-carbon fertilizer alternatives.
iii. The WaPOR project (Donor: Ministry of Foreign Affairs, Netherlands; Lead IWMI; End: 31/10/2026) aims at developing digital advisories and decision support tools that are helping national governments in better targeting and supporting water investments to enhance farm/field level water productivity. In Kenya, WaPOR is leveraging upon the analytics developed under AoW5. The decision support tool aims to identify water use in rainfed and irrigated agriculture and identify the targeting of water investments to enhance agricultural productivity.

7. Plan of Results and Budget

The 2025 Plan of Results and Budget (PORB) of SFP reflects all-inclusive, balanced, and forward-looking investment strategy across all current AoWs, ensuring alignment with the Program's objective to scale inclusive, science-based innovations. Every AoW receives dedicated funding (Table 7.1), demonstrating the Program's commitment to delivering results across technical, institutional, and enabling domains. The PORB represents the allocation of W1-2, and not yet the contribution from W3 and bilateral projects. Please refer to the attached PORB.

AoW8 (Program Performance and Delivery) receives the largest allocation (33%) in the 2025 approved budget, reflecting its pivotal role in ensuring that the Program operates efficiently, accountably, and with measurable results, supporting all AoWs. This AoW includes functions such as coordination of multistakeholder platforms for scaling innovations, partner coordination, MELIA, data infrastructure, analytics and prioritization. By heavily investing in AoW8, the Program ensures that all other AoWs benefit from strong institutional support for delivering results, real-time performance tracking which supports learning loops and adaptive program management.

Among the thematic oriented AoWs, AoW4 (Plant Health and Mycotoxin-Safe Crops) and AoW1 (Climate Change Adaptation and Mitigation) receive a significant portion of the 2025 budget (33% of the total budget combined). These investments reflect the Program's emphasis on managing abiotic and biotic threats that affect farm productivity, food security, and consumer health. The focus on climate-smart and pest-resilient solutions aligns directly with the Program's objective to promote inclusive, sustainable, and profitable whole-farming innovations. AoW2 (Precision Nutrient Management) and AoW3 (Resilient Soils) are also well-resourced (17% of the 2025 total budget). Together, they drive agronomic innovations that are economically viable and environmentally sustainable, including site-specific fertilizer use, nutrient use efficiency, as well as soil health monitoring and improvement. These innovations underpin the Program's objective of sustainable intensification and farm-level profitability.

AoW7 (System Integration through Co-creation), which is the 4th most resourced, highlights the Program's emphasis on participatory design, stakeholder engagement, and contextual relevance of innovations. Given the budget uncertainty, the budget of this AoW was reduced by 50%. This AoW ensures that whole-farm innovation bundles are co-designed with farmers and partners, increasing their significance, ownership, and potential for sustained uptake. Its early resourcing creates the path that responds to the diversity of farming systems and socio-economic factors.

AoW5 (Integrated Water Management) and AoW6 (Scale-appropriate Mechanization), while modestly resourced relative to other AoWs, play pivotal roles in enabling and amplifying the impact of innovation bundles. AoW5 focuses on addressing water-related challenges through improved water management practices, while AoW6 promotes the validation and delivery of appropriate mechanization options as a strategic pathway for scaling innovations and reducing labor burdens, especially among women. These interventions support more stable and productive cropping systems, a priority for SFP.

AoW0 (Crosscutting and Management) is funded to deliver key enabling functions including executive oversight and stewardship, strategic communications, programmatic MELIA functions, and financial management. These investments, along with those in AoW8, support program-wide coherence, integration across AoWs, and enhanced responsiveness to partners and beneficiaries and provide common governance and digital tools.

The AoWs budget targets a foundational set of prioritized HLOs, several of which focus on developing, validating and bundling innovations for scaling, developing innovation specific training materials and models, building platforms and co-creation processes, including operationalizing multi-stakeholder platforms and harmonized digital advisory systems, data analytics and infrastructure. These outputs aim at generating early evidence on agronomic performance and are essential to ensure that scaling is coordinated, participatory, and adapted to local needs. These HLOs ensure that the Program does not just deliver outputs but creates systems, partnerships, and evidence that will shape and accelerate delivery from 2026 onward. These early investments support systemic readiness, allowing subsequent scaling of validated innovations efficiently and inclusively.

In moving beyond continuity, the Program deliberately allocates part of the 2025 budget to a "Pause and Reflect" session with stakeholders, to identify emerging priorities and opportunities. 2026 budgets will therefore be adjusted based on such issues where relevant. This opportunity will be essential for the evolution from the current AoW to the future AoW within SFP.

The distribution of budget across Centers mirrors their roles within specific AoWs and regions. The distribution supports technical and appropriate expertise, also enabling collaborative delivery of common outcomes. The initial baseline budget confirmed in February 2025 (US$36.6M) was reduced to US$26.6 in May, and further reduced to US$25.1M in June. The Program leadership team reached an agreement to have a higher reduction of budget (50%) in AoW 7, and share proportionally across all centers the budget reduction allowing some flexibility within AoW budgeting.

Table 7.1 **2025 Budget Allocation Across Centers and AoWs**

	SFP Updated Budget Allocation (US$)													
	SO	AfricaRice	Bioversity	CIAT	CIMMYT	CIP	ICARDA	ICRISAT	IFPRI	IITA	ILRI	IRRI	IWMI	TOTAL
Inception funding, Cross-cutting	-	-	-	-	-	221,648	-	-	-	377,958		-	-	**599,606**
Cross-cutting & Program Management	184,200	26,250	-	52,500	52,500	52,050	26,250	26,250	-	334,678	-	78,750	26,250	**859,678**
AoW 1	-	34,245	240,193	339,986	579,729	157,684	261,699	63,722		1,018,263	46,826	151,393	266,849	**3,160,589**
AoW 2	-	129,077	28,903	595,755	321,690	138,376	88,127	105,604	-	843,551	-	72,499	19,755	**2,343,338**
AoW 3	-	135,004	-	343,105	416,836	112,652	96,112	109,795	-	233,600	36,736	58,494	45,222	**1,587,556**
AoW 4	-	305,120	311,952	343,105	1,157,000	530,090	274,131	105,673	31,103	1,024,937	54,585	439,067	-	**4,576,764**
AoW 5	-	19,757	-	196,506	51,074	94,735	55,690	37,195	-	65,006	36,736	33,676	412,726	**1,003,100**
AoW 6	-	79,027	-	-	260,623	60,337	50,358	38,568	-	33,269	75,176	228,555	-	**825,913**
AoW 7	-	6,389	341,852	271,365	509,294	63,585	209,180	93,294	-	550,120	126,821	147,100	5,913	**2,324,914**
AoW 8	-	585,990	73,752	1,029,315	1,197,237	250,325	276,728	188,279	537,025	2,786,680	112,748	533,822	252,564	**7,824,465**
Total (USD)	**184,200**	**1,320,858**	**996,652**	**3,171,637**	**4,545,985**	**1,681,482**	**1,338,278**	**768,380**	**568,128**	**7,268,062**	**489,628**	**1,743,356**	**1,029,279**	**25,105,923**
% variation compared to the $26.6M scenario		-6.04%	-6.04%	-6.04%	-6.04%	-6.04%	-6.04%	-6.04%	-6.04%	-6.04%	-6.04%	-6.04%	-6.04%	

8. Cross-Portfolio linkages and geographic coordination

8.1 Cross-portfolio linkages

Collaboration among Programs and Accelerators is expected to happen by co-financing, co-locating and co-implementing activities. Although the budget uncertainty has limited these interactions, we are providing examples of collaborative activities below.

The collaboration between SFP and Multifunctional Landscapes (MFLs) focuses on integrating conservation-focused agricultural practices at the farm level with broader landscape-scale water management strategies. Currently being developed in Kenya and Ethiopia through bilateral funding, this strategy framework aims to expand to other regions. Key areas include improving water and soil management from farm to landscape level (SFP AoWs 3, 5 and 7)—enhancing agronomic gains through nutrient efficiency while integrating agroecological and regenerative approaches (MFL). Another focus is prioritizing water management solutions across scales, ensuring that landscape-level water improvements translate into farm-level productivity (SFP AoWs 1 and 5, MFL AoWs 1 and 2). Interactions and feedback across scales will be analyzed using tools developed under MFL AoW 6 and SFP AoW 5.

The collaboration between SFP and the Digital Transformation Accelerator (DTA) is primarily facilitated through AoW8, focusing on data standardization and AI-driven research advancements. A key initiative involves integrating agronomy and breeding data into DTA's "AgPile" to enhance accessibility across CGIAR and partner databases. Additionally, discussions are underway to explore applications within DTA's AoW2, leveraging Large Language Models (LLMs) and AI-driven insights for research and scaling. SFP has already contributed human-verified question-answer pairs to support AI initiatives like AgriLLM, currently in development at the DTA Hub in Abu Dhabi, and partners such as Digital Green. Another emerging collaboration centers on the Multi-Dimensional Inclusivity Index (MDII), designed to assess digital inclusion, with SFP providing use cases to test, validate, and refine the framework.

SFP and Climate Action collaborate on climate analytics and scaling strategies for low-emissions, climate-resilient food, land, and water systems. SFP focuses on field- and farm-level solutions, which inform the broader climate policy and investment strategies developed by Climate Action. Additionally, SFP leverages climate data—hazard projections, localized forecasts, and measurement frameworks—provided by Climate Action to refine interventions and assess their impact. The programs also coordinate efforts on drought management and climate shocks, with Climate Action developing general frameworks and forecasts, guided by SFP's behavior change strategies and practice options. Shared staff, data and analytics, co-developed outputs, and joint engagement with partners such as national governments and IFIs ensure alignment and integration across initiatives.

SFP AoW3 is collaborating with the Policy Innovation Program (PIP) to advance fertilizer and soil health efforts across Africa. Plans under discussion include a joint survey in Kenya on fertilizer subsidies and soil health, with SFP potentially leading the soil testing component and contributing to the study and dissemination. In Nigeria or Malawi, the programs are exploring ways to align efforts on fertilizer policy and soil health, though specifics are still being defined. Additionally, discussions are underway to co-convene an Africa-wide conference on fertilizer and soil health, supporting the goals of the Nairobi Declaration. The two programs are increasingly aligning, as reflected in their joint contributions to upcoming conferences.

SFP and the Scaling for Impact Program are co-designing and evaluating scaling pathways for innovation bundles and digital tools, contributing to the SFP scaling strategy. This Program fosters synergies across CGIAR and its partners, ensuring scalable, inclusive, and responsible innovation delivery. The collaboration is progressing well, with teams mapped across both programs and efforts underway to strengthen coordination. Notably, Scaling for Impact team members participated in the PAASS framing workshop in February 2025 in Nairobi and have been invited to peer review and refine the PAASS guidelines and playbook.

8.2 Geographic coordination

SFP has established the PAASS partnership platform to foster collaboration among key stakeholders—including CGIAR Centers, Programs/Accelerators, NARES, national governments, farmer collectives, private sector actors, and development organizations—ensuring alignment on agricultural innovation and scaling. PAASS serves as SFP's scaling framework, built on lessons from three Initiatives (EiA, PHI and MFS). Set for finalization and implementation in 2025, it will guide innovation scaling through close collaboration with the Scaling for Impact Program (S4I). This approach facilitates geographic coordination, enabling stakeholders to identify priority constraints and drive scaling efforts efficiently. By integrating multiple Programs and Accelerators within the same or neighbouring regions, PAASS enhances co-design and validation of innovations within relevant environmental, economic, and social contexts. This structured approach minimizes duplication, improves resource efficiency, and strengthens shared logistics, infrastructure, and knowledge systems.

PAASS enhances efficiency through joint implementation, reducing costs via shared logistics and pooled resources. It accelerates the research-to-impact pipeline by ensuring that solutions co-developed with local stakeholders are relevant, adopted, and sustained. Farmer collectives provide field insights, while researchers contribute scientific expertise, creating innovations that are both practical and evidence based. The platform fosters open knowledge exchange, strengthening adaptive capacity and responsiveness. Crucially, PAASS aligns with stakeholder priorities—national governments gain demand-driven innovations supporting agricultural strategies, donors see higher returns through scalable impact, and local partners, including NGOs and the private sector, engage more effectively in co-design. Most importantly, smallholder farmers and their communities receive solutions tailored to their needs, backed by mechanisms for shared ownership and long-term sustainability.

9. Risk Management

Please refer to the following ***link*** with the Program/Accelerator's full risk register. This link is open to ISDC colleagues and others will be granted access, on request.

#	Category	Title	Description	Current Risk Level	Target Risk Level	Actions /Controls to manage risks
1	**Partners and Partnerships**	Uncontrolled change causing disruption	Uncontrolled changes emerging from transitioning from Initiatives to Programs can disrupt Center and partner teams' key research activities, resulting in losing progress in the Program AoWs, momentum and funding.	16	12	The Program Leadership Team involving interim Directors and AoW leaders are meeting frequently for planning and reporting 2025 deliverables, building on progress made by the Initiatives. AoW leaders in turn are involving Center teams in the discussions and implementation of activities
2	**Talent**	Failure to attract, and/or retain talent	Uncertainty and limited clarity of roles, responsibilities and opportunities for Center and partner staff in the Program structure can result in limited attraction and retention of key talent, weakening the Program research and scaling teams.	20	15	The Program Leadership Team, particularly AoW leaders are discussing with Center teams the contribution of the staff to the AoW deliverables, so that key subject matter specialists are retained as members of SFP.
3	**Partners and Partnerships**	Limited inter-Program integration	Speed of change, uncertainty of financial resources and limited system approaches would result in insufficient inter-Program integration/collaboration, reducing synergies key for Sustainable Farming system integrated approaches, outputs and outcomes.	20	15	The Program Leadership Team is discussing potential collaborative activities with other Programs and Accelerators. Priority collaborations will evolve those conversations into concrete actions that can start to be co-financed, co-located and co-implemented in 2025, and particularly emphasized in the 2026-2030 period.
4	**Partners and Partnerships**	Limited stakeholder involvement and potential conflicts	Limited stakeholder involvement in Program planning and implementation due to speed of change, number of partners, and funding uncertainty can lead to misalignment of goals, conflicts, delays and/or non-delivery of Program outputs and outcomes.	16	12	The Program Leadership Team is planning to involve partnership platforms (created during Initiative implementation) in further discussions for prioritization of locations and research topics to be conducted by the Sustainable Farming Program team. This consultation will be essential for reorienting the Areas of Work of the Program.

5	**Funding**	Funding uncertainty limits proper planning	Funding uncertainty limits proper planning of research and scaling activities by Center and partners' teams, delaying and/or interrupting the delivery of Program outputs and outcomes.	20	15	(1) Connect with funders to gain visibility on other risk arising for their side; (2) The team is conducting budget scenario planning considering 25% and 50% likely reduction, which will inform prioritization of deliverables for 2025 and 2026-2030 plan.

10. Addressing feedback on select topics

Concise summary of feedback	**Details on how the Program/ Accelerator has addressed or will address this feedback** (in the Inception Phase or beyond)
There is a lack of information on how water systems are covered in the Portfolio.	SFP addresses water systems through two AoWs: AoW1 (Climate Adaptation and Mitigation on Farms) and AoW5 (Integrated Water Management), each with distinct outputs and deliverables. In future AoW configurations, this research will be integrated into areas focused on shock preparedness and addressing farm binding constraints.
There is a lack of information on how climate mitigation is covered in the Portfolio.	SFP's AoW1 (Climate Adaptation and Mitigation on Farms) focuses on climate mitigation at the farm level. Moving forward, this work will transition into a new AoW dedicated to enhancing preparedness and enabling rapid responses to emerging biotic and abiotic farm shocks.
There is a lack of information on how tradeoffs and synergies in mixed systems are covered in the Portfolio.	SFP’s AoW7 (System Integration through Co-Creation) focuses on analyzing trade-offs in sustainable farm management. Moving forward, this work will continue to evaluate trade-offs of potential solutions for farm binding constraints within the future AoW configuration.
There is a lack of information on how environmental health and biodiversity is covered in the Portfolio.	SFP integrates environmental topics and indicators, including GHG emissions, into the MELIA system, particularly within its AoW1 framework. Monitoring these indicators will be essential for evaluating strategies to enhance preparedness for shocks and address binding constraints in future AoW configuration.
There is a lack of information on how orphan and opportunity crops and pulses are covered in the Portfolio.	SFP’s AoW3 on resilient soils focuses on key practices such as diversification, crop rotation, and integrated soil management, including the use of legumes. As part of the upcoming prioritization process, there is potential to broaden this framework by incorporating orphan and opportunity crops, further strengthening soil health and system resilience.
There is a lack of information on how youth and social inclusion (that does not relate to gender) is covered in the Portfolio.	SFP considers the implications of job creation in service-related sectors, particularly through AoW6 (Scale-Appropriate Mechanization). All solutions developed by SFP AoW will undergo assessment for gender and youth impact. To support this effort, a multi-Center, cross-AoW gender working group has been established, actively guiding and strengthening the work. For digital solutions, SFP is currently testing the multi-dimensional inclusivity index as a suitable tool for evaluating the digital inclusiveness of the products developed as part of the MELIA.

Concise summary of feedback	**Details on how the Program/ Accelerator has addressed or will address this feedback** (in the Inception Phase or beyond)
There is a lack of information on how capacity sharing (as it relates to internal learning and decolonization of research) is covered in the Portfolio.	SFP adopts capacity sharing as a core intervention principle, with specific deliverables embedded within each AoW. AoW8 (Program Efficiency and Delivery) aims at delivering more integrated capacity-sharing mechanisms. This work will remain a fundamental part of the new AoW configuration, as capacity sharing will be essential for leveraging recommendations from advisory services, strengthening stakeholder preparedness for biotic and abiotic shocks, facilitating the adoption of solutions to binding constraints, incubating new research ideas, and advancing data science, MELIA, and partnership approaches.

Appendix 1: Addressing specific feedback

Concise summary of feedback	**Details on how the Program/Accelerator has or will address this feedback** (in the Inception phase or beyond)
An area needing further consideration is how the proposal describes highly diverse farmers and farming practices as relatively homogenous and in need of supply-side solutions from experts. While there is some mention of codesign and the role of farmer-facing services and support organizations as the primary mechanism for effecting change, many entries do not sufficiently emphasize farmers (and their service suppliers) as real partners. Additionally, there is an assumption that farmer-facing services will readily adopt new bundles and technologies, but this needs further unpacking and detailing.	The principle of co-creation with farmers and other key stakeholders will be emphasized during the elaboration of annual work plans. Methodologies to determine farmer typology to design targeted interventions will be used to adequately address the needs of different types of farmers. Co-creation will be particularly monitored by AoW 7, which will emphasize farmer participation. Partnership platforms (PAAS) will identify demand partners and their needs to co-create and co-develop solutions in an integrated manner, which is a priority for AoW8.
While the research design is supported by relevant literature showing evidence of social and gender inequalities in agri-food systems, the inclusion and gender-responsive approach appears diffuse in the proposal and greater details on incorporating this issue is needed in many Areas of Work. The main strategy to address these issues is linking with the Social and Gender Inclusion Accelerator, which includes team specialists to support this work at the Area of Work level.	AoW7 has started during the inception phase to specifically include women and youth demands for prioritizing, planning, conducting and scaling bundles of innovations. A Gender Working Group across AoW and Centers has been initiated, which is helping all AoW to perform gender responsive research and coordinate their actions with the Gender Accelerator.
Research questions could be more descriptive regarding the social and economic factors that impact farmers' ability to use and deploy existing and developed knowledge.	Co-creation methods combined with system sciences will be used to address social and economic factors that influence development and scaling of agricultural innovations, particularly in AoWs7 and 8. AoW research questions and outputs have been revised to emphasize social and economic factors where relevant.
New geographies and emerging pests should be given due consideration when prioritizing activities during the Inception Phase.	AoW4 is addressing plant health challenges and is supporting national partners to monitor both existing and emerging pests and strengthen diagnostics, surveillance, predictive modelling and risk analysis.
There is limited discussion on how existing farming systems were conceived and specific proposed outcomes were determined. Including a comparison of work aims with	The definition of outcomes took into consideration the lessons learned, and progress made by Initiatives (for example, the work on farming system definitions conducted

those achieved in the previous cycle of Initiatives would be beneficial.	as part of the EiA and MFS Initiatives). Outcomes are being revisited (for relevance and adding measurable indicators) during the Inception Phase.
AoW2: Additional information on how AgWise was used and its impact in previous Initiatives would strengthen this section.	During the Inception Phase, a more detailed analysis of progress made by AgWise is being carried out to build on its strengths and address weaknesses to improve targeted fertilizer recommendations.
AoW3: This Area of Work is high on ambition but low on details and problematization. Aiming to deliver healthier soils with more organic matter across a large area in the semi-arid and humid tropics, where SOM content is low/variable and SOM oxidation is high, will be incredibly difficult. More spatial consideration is needed in this Area of Work. This Area of Work needs greater details on actions that will be taken to improve soil health. Since this Area of Work will have less existing information and expertise to draw on, additional details on key research partners, methods, and models developed elsewhere would be helpful. A comparative advantage analysis is needed to clarify why this work should be pursued by CGIAR.	AoW3 is addressing these recommendations in the revised 2025 work plan for 2025, emphasizing a stronger problem orientation. It included an analysis of prevalent challenges at various levels (e.g., geography and farm typology, political and socio-economic barriers, soil health assessment and monitoring, R&D) that may limit the scaling of solutions that address both agronomy and soil health increase. Additionally, an updated comparative analysis was carried out to involve key partners that can contribute to or take the lead in developing some of the suggested solutions. This work will continue in the new AoWs 1 to 3 that will address soil-related constraints.
AoW4: This area of work would be strengthened with a plan on how it might respond to emerging pests (e.g., a new disease outbreak).	AoW 4 includes activities and outputs focused on developing state-of-the-art and cost-effective tools to enhance surveillance, predictions and early warning systems as well as capacity sharing for NPPOs and NARES to enhance their detection, diagnosis, surveillance, monitoring and risk assessment capabilities for better preparedness. This will include specific plans for rapid response to emerging pests
AoW5: It is unclear why most proposed work for non-irrigated environments is not part of AoW 1 and/or AoW 7, given the Program’s emphasis on integration. Most interventions for water management will likely include extensive crop-related agronomic recommendations. The details of appropriate water stewardship and management are thin and idealistic.	Strong inter-AoW interactions are planned. AoW1 will integrate solutions that enhance adaptation to climate change from other AoWs (e.g. AoW5). Collaborative activities with be conducted with other Programs and Accelerators, such as with the Multifunctional Landscapes and Climate Action Programs to address water-related challenges at farm- and landscape-levels.
AoW6: The heading for Section 6.6.1 does not adequately describe the proposed activities. It is unclear whether this Area of Work will	AoW6 flagship product for 2025, the Global SAM Dashboard, will offer country-level mechanization status, gap analysis for key

develop new SAMs, adapt existing ones, or simply write SOPs for them. There should be a clear link between the SAMs developed by the program, their impact at the farm level, and the factors inhibiting their adoption. If the development of post-harvest equipment presents a significant opportunity, it is unclear why it should be excluded at this stage of the program's development.	production systems, and suitability maps. Two key supporting tools—the SAM Options Inventory (covering field and post-harvest tools) and the Service Provision Models Inventory—will further enhance accessibility and decision-making. Activities will focus on developing and adapting SAMs to meet regional demands, with post-harvest mechanization fully integrated into planning to reduce losses and strengthen value chains across diverse agricultural settings.
AoW7: There is good attention to upscaling and cocreation, but more could be done in this regard as mentioned elsewhere in this review. The linkages between this activity and other AoWs could be strengthened, and the proposal should clarify how this Area of Work differs from Area of Work 1.	AoW7 aims at developing co-creation methods and modelling system approaches (combining biophysical, socioeconomic and behavioural factors) for bundling farm innovations. Unfortunately, since this AoW was supposed to be financed by USAID contributions, the team has had to deprioritize the 2025 deliverables and reduce the budget by 50% to adjust to the budget reality updated in May 2025.
AoW 8: This Area of Work should have its own evaluation indicators to avoid dilution within other Areas of Work.	AoW8 is playing an integrative function across all other AoW through the development of a digitally enabled MELIA system, data infrastructure, novel capacity-sharing models, multi-stakeholder platforms and integrated policy related information. During the Inception Phase, all AoWs are developing more specific performance indicators, with AoW8 also developing indicators to ensure that its contributions will be clearly tracked and documented while also preventing the risk of being subsumed in other AoWs. These indicators will focus on enhancing Program efficiency for facilitating integrative research, data management and scaling integrated solutions, with emphasis on digital solutions.
The sensitive issue of the willingness to share data among partners is not fully addressed nor is the special effort needed to adapt integrated management strategies to different geographies, climates, or cultures at the farm level.	Integrated data management requires addressing the need for culture change, capacitation, value demonstration and effective tools to address the sensitivity of data sharing among partners. AoW8 is building on these elements and progress made during Initiative implementation, particularly under EiA, and with the DTA. Key efforts and deliverable include those around data standards, management, and harmonization; digital tools; governance; capacitation; and enabling environment among CGIAR Centers

	and their partners. These efforts are being emphasized during the elaboration of annual work plans and identification of collaborative activities with partners.
The Program clearly describes how innovative collaborative approaches, mechanisms for establishing partnerships, cooperation with other programs and accelerators, and building on activities of previous initiatives can lead to significant impact. The activities of CGIAR in Ethiopia are used as a compelling example. It is less clear how these collaborative, integrated research efforts will be undertaken in countries where CGIAR has a limited presence. The link with other Programs and accelerators to achieve effective scaling and impact is explicit.	As part of SFP inception, there is emphasis on robust and agile partnership platforms through a strengthened mechanism (PAASS) that is a cornerstone for collaborative and integrative efforts, with partners, CGIAR Programs and Accelerators and other projects (as indicated above). Where CGIAR presence is limited, digital presence will be explored (e.g., for capacitation, with contextual and targeted modules available online, that can be delivered as interactive online courses, or in audio, video, or factsheet/document format. Links with other Programs, particularly the Scaling for Impact Program and the Capacity Sharing Accelerator will also help in coordinating work in countries with CGIAR limited presence.
The role of farmers in MELIA needs to be highlighted, as does the impact of activities on non-technical outputs such as institutional strengthening and community infrastructure. The robust Capacity Sharing plan should ensure uniformity in approaches and data collection across partners. Impact assessment should be designed to mitigate the risk of stakeholders' unwillingness to share data.	During the development of the MELIA plan, the role of farmers will be specified and highlighted, and indicators related to non-technical aspects included. The capacity sharing plans (at the level of AoW 8 and at Program level) will include activities such as strengthening data management capabilities, encouraging best practices through value demonstrations (addressing "what's in it for me"), deploying data sharing agreements across partners and implementing strategies to mitigate the risk of stakeholders not being willing to share data.